

RENSSELAER MECHATRONICS

Realistic DC Motor Step Response Simulation

Objectives:

- Compare 1st and 2nd order DC motor models, with and without friction
- Examine the hardware effects of discretization, delay
- Compare the simulated response with the experimental response

Parts 1 and 2 are simulation only – no hardware is needed. Simulink is used to solve equations of motion (Ordinary Differential Equations or ODEs) and simulate/predict the response of the system with different nonlinearities. The simulated response is verified at the end with an experiment.

Part 1: 1st and 2nd order Step Response

Objectives:

- Compare 1st and 2nd order DC motor models, with and without friction
- Determine what model is a “good enough” approximation to use
- Evaluate if the simulation matches the experiment step response

Background Information:

The 2nd order transfer function for a DC motor is:

$$G(s) = \frac{\omega(s)}{V(s)} = \frac{K_t}{LJs^2 + (JR + Lb)s + (Rb + K_tK_b)}$$

If the inductance is “small” it can be neglected ($L=0$) and the first order transfer function is obtained:

$$G(s) = \frac{\omega(s)}{V(s)} = \frac{K_t}{JR s + (Rb + K_tK_b)}$$

Where the motor parameters are:

- K_t – torque constant
- K_b – back emf constant

- b – viscous damping coefficient
- R – armature resistance
- J – armature resistance (or combined motor and load inertia if load is attached)
- L – inductance

This lab will examine the effects of the inductance and damping terms. It will also examine the effect of nonlinear coulomb friction (which cannot be included in a transfer function)

$$F_c = \text{sign}(\omega) * T_f$$

Where F_c is the total coulomb friction torque – it is a constant magnitude T_f that depends on the direction of rotation $\text{sign}(\omega)$. This nonlinear effect will be included in the simulation.

Define Important Motor parameters:

Create an m-file that will have all the important motor parameters. Use the parameters below. These should be close to the parameters you found in the Parameter ID Lab.

```
% Motor parameters:
Rm = 13;           % ohms
Kb = 0.155;       % Vs/rad
Kt = 0.1;         % Nm/A
Bm = 0.0002;      % Nms/rad (viscous friction)
Lm = 0.004711;    % H (estimated)
Jm = 0.00005;     % kgm^2 (combined J)
Tf = 0.002;       % Nm (coulomb friction)
encoder_counts=1400; % number of counts (if using quad encoding)
Vsupply=4.65;     % driver supply voltage (max saturation voltage)
```

Figure 1: Motor parameters stored in script "Motor_Parameters.m"

Simulation Diagram

Create the following simulation diagram that has 3 different DC motor models, a 1st order with no friction, a 1st order with viscous friction, a 1st order with viscous + Coulomb/Dry friction and a 2nd order.

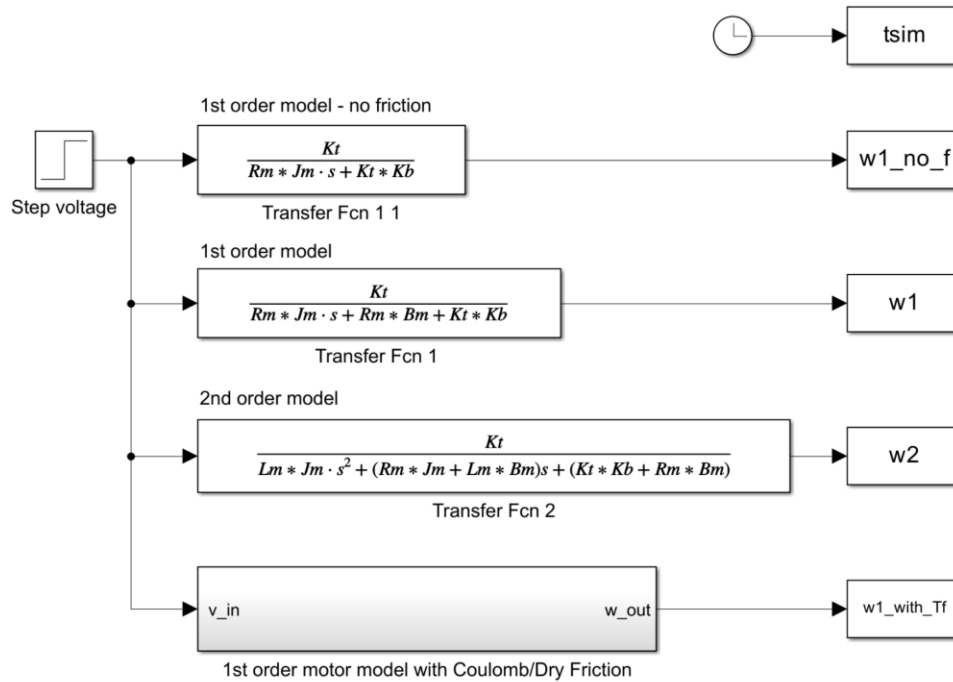


Figure 2: Simulink diagram for 4 different motor models

The three blocks use transfer function blocks, the one is a subsystem you will create as shown below.

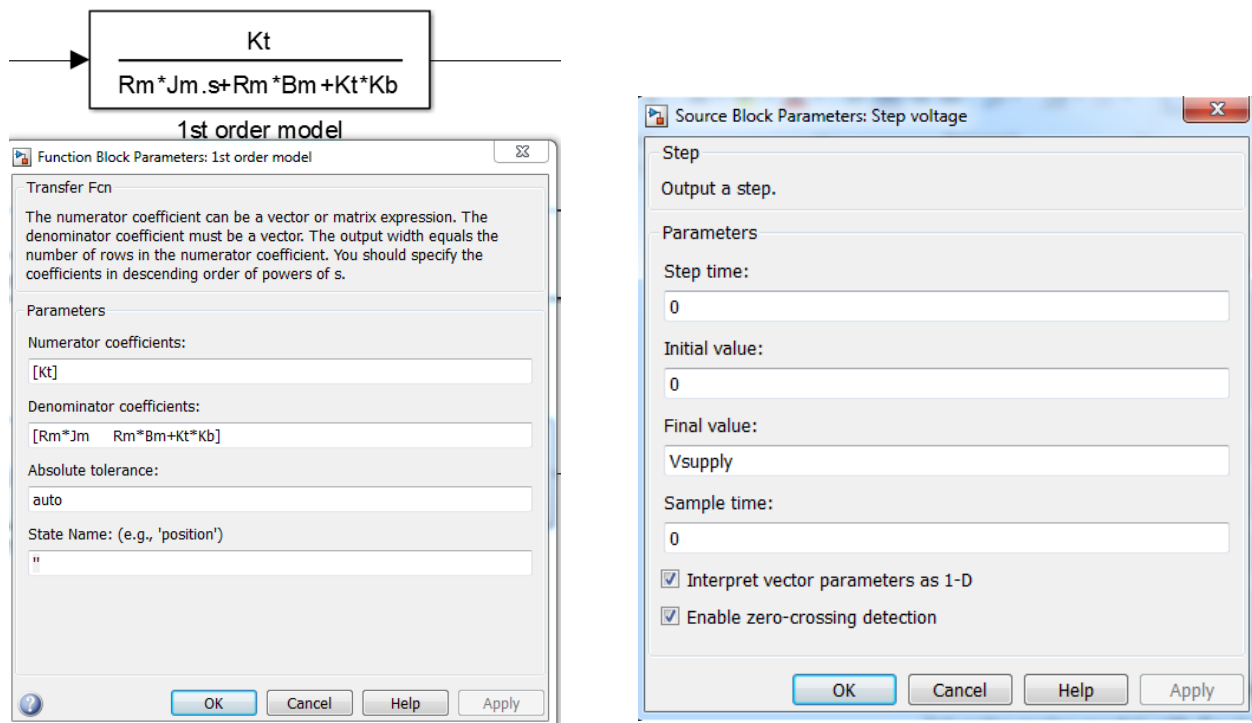


Figure 3: Transfer function and Step block parameters

Be sure to change the initial step time to zero.

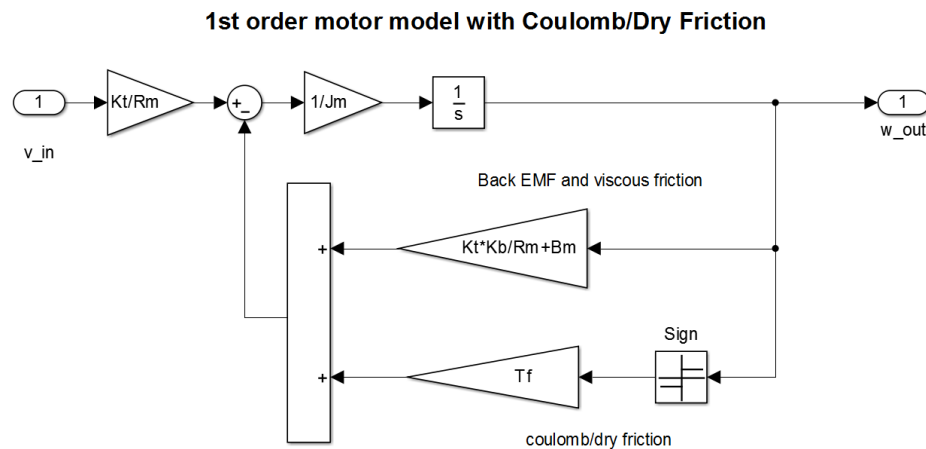


Figure 4: The subsystem contents of the previous figure

Simulation Script

Create an m-file that will contain all the important simulation settings, run the simulations and plot the results:

```
clc, clear all, close all

% load motor parameters:
Motor_Parameters_MinSegMicro

% simulation parameters
Tsim_final=.3;           % final simulation time in seconds
sim_step_size = .0001; % how often we want data printed from the simulation

% Simulate the model
sim('Motor_Step_Response_Linear_1st_2nd_Coulomb')

% plot the response in RPM
figure(1), hold on
plot(tsim, w1_no_f, 'k');
plot(tsim, w1, 'b');
plot(tsim, w2, 'r--');
plot(tsim, w1_with_Tf, 'g');

xlabel('Time (seconds)')
ylabel(['angular velocity rad/sec. - input ', num2str(Vsupply), ' v step'])

legend({'1st order - no friction',...
       '1st order - viscous only',...
       '2st order',...
       '1st order - viscous + coulomb'})
```

Figure 5: m-file "Sim_Motor_Response.m" used to simulate and plot motor model simulations

Use the following simulation solver settings:

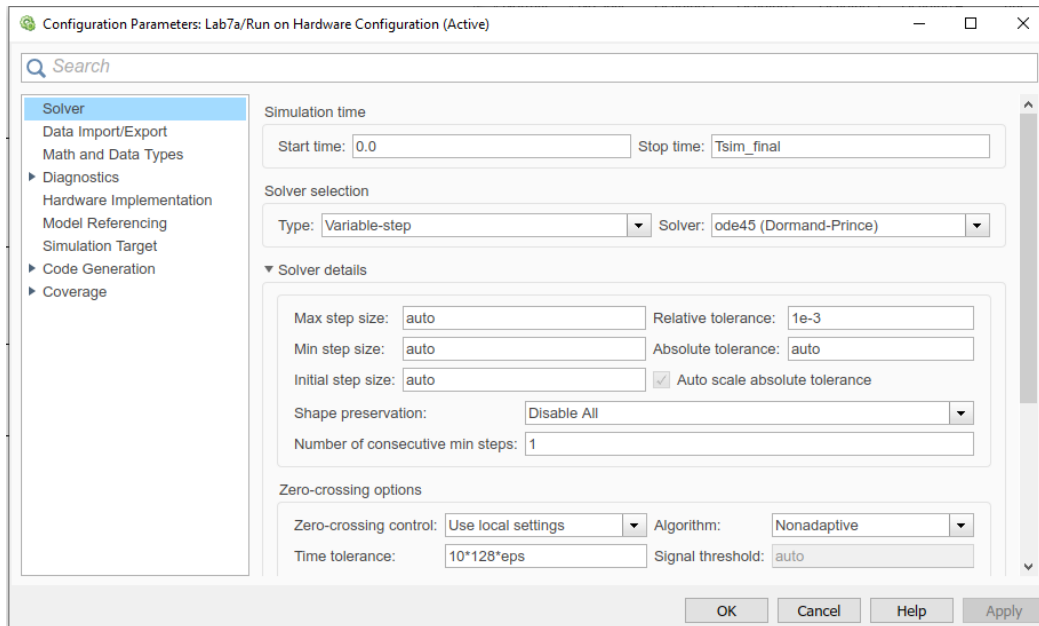


Figure 6: Simulation settings used to solve system response

- Notice the different settings here are used to **solve** equations of motion, **not** to program hardware
- Notice the data recorded by the simulation is specified in the “To Workspace” Blocks
- The **parameter `sim_step_size=.0001 sec`** is how often the **simulation** will provide you data points as it solves the system response. Notice it is much smaller than the step size used to program the hardware (typically .005-.03 sec). This is to ensure your calculated predicted solution from Simulink has enough resolution to capture all dynamics that may occur in the hardware system.
- The size of the voltage step is V_{supply}

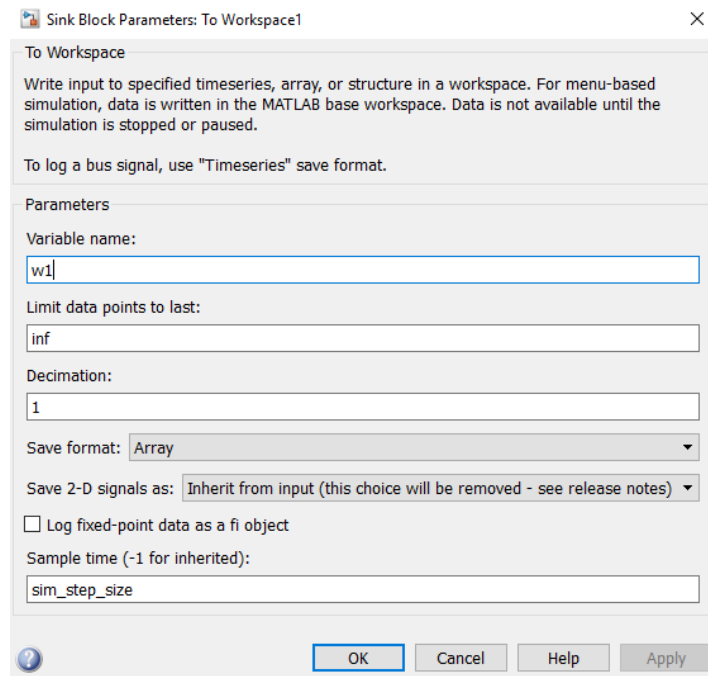


Figure 7: settings for "To Workspace" blocks

Run the simulations as specified in the m-file and plot the results.

Questions:

- For the specified set of parameters
 - Can inductance in the system be neglected?
 - Compared to the first order model with viscous friction only – what percent error does the final speed have for the model with no friction?
 - Compared to the first order model with viscous friction only – what percent error does the final speed have for the model with viscous and coulomb friction?
 - Does this step response match the experimental step response from previous labs?

Part 2: 1st Order System with Discretization and Delay

Objectives:

- Add the hardware implementation effects to the simulation
 - Discretization - from sensors and data type
 - Delay - from sampling data

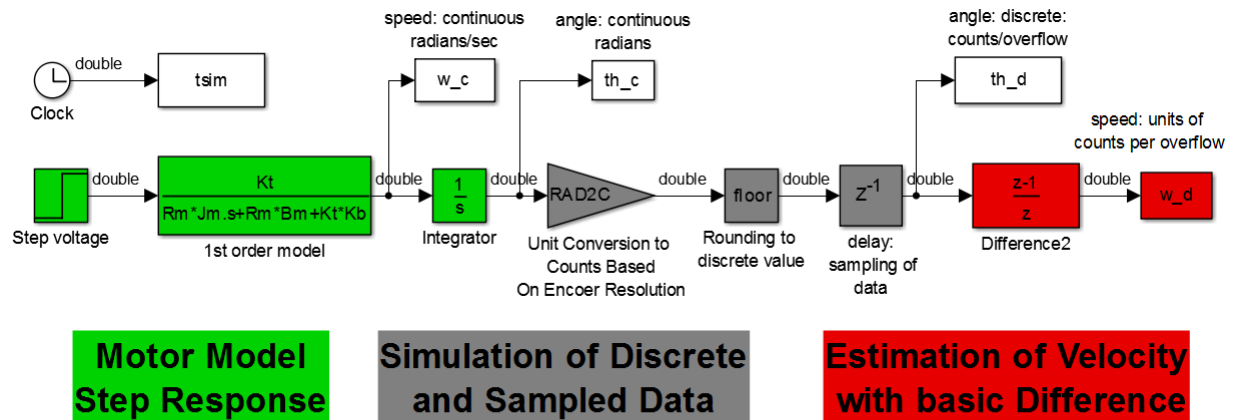
Background Information:

The implementation on the hardware introduces several effects:

- Encoder provides only discrete data points – 1400 counts per revolution
- The hardware only samples this data at specific time interval T_S – a delay introduced from measurement to measurement
- The velocity is calculated from the difference between two discrete position measurements

Simulation Diagram

Create the following simulation diagram:



Notes:

- This simulates the discrete nature of the encoder and the sampling of the data.
- All the data types are doubles & all the math is floating point: this does not take into account data types and fixed point math!

The delay block has a sample time specified by "TS" our microcontroller sample time,

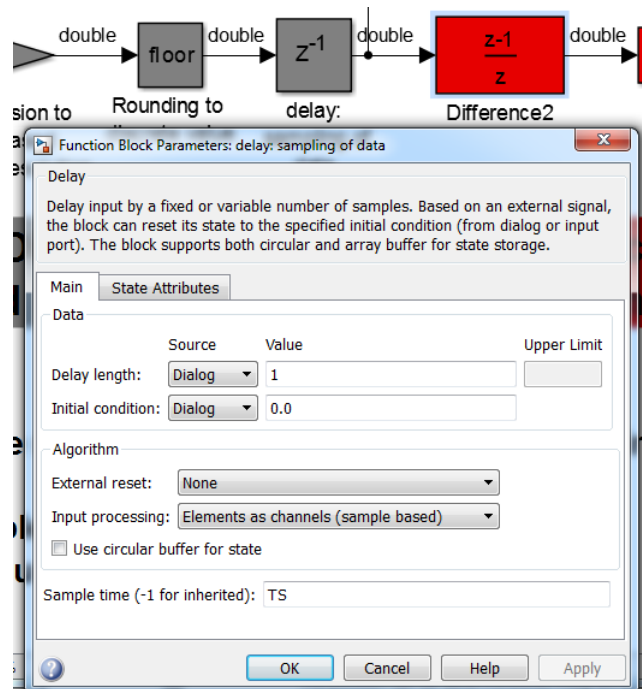


Figure 8: Delay block setting

Diagram Overview:

- Step response is simulated with the first order model with friction (transfer function block)
- Integrator computes the position (what is actually measured in the system)
- Position is converted to counts by the number of encoder counts available in quadrature mode – this represents the discrete resolution of your measurement system
- Counts are rounded to integers to represent the discrete data type of the encoder (int16)
- Delay represents the delay time in measuring the signal with the microcontroller
- And the difference is the calculation of the speed as implemented on the microcontroller

The m-file can be run to simulate and plot the data:

```
%% 1st order system with discretization and delay and velocity calculation
figure(2)
Tsim_final=.5;           % final simulation time in seconds
TS=.03;sim_step_size = TS;
sim('Motor_Step_Response_Linear_With_Encoder_And_Delay')
plot(tsim, w_c, 'b'), hold on
plot(tsim, w_d*C2RAD/TS, 'g')

TS=.003;sim_step_size = TS;
sim('Motor_Step_Response_Linear_With_Encoder_And_Delay')
plot(tsim, w_d*C2RAD/TS, 'm')
```

Notice that the “sim_step_size” is set to “TS”. This is forcing the simulated solution to provide data points at the same frequency that the hardware obtains data points.

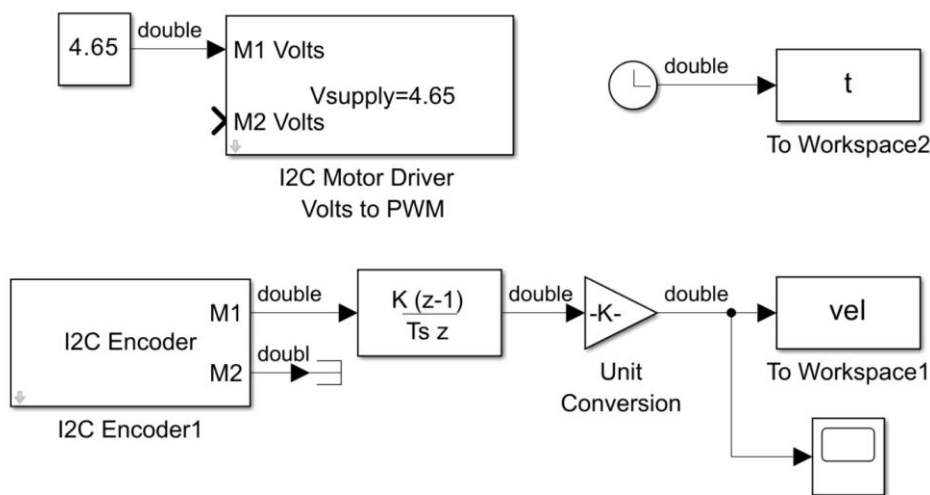
Questions:

- Provide a plot of the 1st order system simulated response of the discretized system at TS=0.03 seconds and the discretized system at TS=0.003 seconds

Part 3: Experimental Response

Experimental Response

Use the following Simulink diagram (used previously) to capture the step response in at a time step of 0.03 seconds and 0.003 seconds. (you may have this already from previous labs)



- Use the Encoder block for your system
- Make sure the drive voltage switch is set to use VIN (usb voltage)
- Be sure “enable override detection” is enabled on pin 13 to ensure your program does not overflow

Questions:

- Provide a plot the of the experimental step response versus the simulated discretized step response for both 0.03 seconds and 0.003 seconds (be sure to convert your units correctly!)
- How well does your simulated response match the experimental response now?